Propositional rules: Axioms, basic actions, conditionals, switching, composition.

$$\bot \langle \boldsymbol{a} \mid - \rangle P \qquad P \langle - \mid \boldsymbol{\alpha} \rangle \top \qquad P \langle \lambda \boldsymbol{x}.\boldsymbol{x} \mid \lambda \boldsymbol{x}, \boldsymbol{v}.\boldsymbol{v} \rangle P \qquad \frac{P_\exists \to Q_\forall \in \mathrm{Ax}}{P_\exists \langle - \mid - \rangle Q_\forall} \qquad \frac{P_\exists \langle - \mid - \rangle Q_\forall}{P'_\exists \langle - \mid - \rangle Q'_\forall} \text{ for } \frac{P_\exists \to Q_\forall}{P'_\exists \to Q'_\forall} \in \mathrm{Rule}$$

$$\frac{P \langle \boldsymbol{a}, \boldsymbol{b} \mid \boldsymbol{\alpha} \rangle Q \wedge R}{P \langle \boldsymbol{b}, \boldsymbol{a} \mid \tilde{\boldsymbol{\alpha}} \rangle R \wedge Q}\, {}^{p\wedge_R} \qquad \frac{P \wedge Q \langle \boldsymbol{a} \mid \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle R}{Q \wedge P \langle \tilde{\boldsymbol{a}} \mid \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\alpha}} \rangle R}\, {}^{p\wedge_L} \qquad \frac{P \langle \boldsymbol{a}, \boldsymbol{b} \mid \boldsymbol{\alpha} \rangle Q \vee_c R}{P \langle \boldsymbol{b}, \boldsymbol{a} \mid \tilde{\boldsymbol{\alpha}} \rangle R \vee_{\tilde{c}} Q}\, {}^{p\vee_R} \qquad \frac{P \vee_c Q \langle \boldsymbol{a} \mid \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle R}{Q \vee_{\tilde{c}} P \langle \tilde{\boldsymbol{a}} \mid \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\alpha}} \rangle R}\, {}^{p\vee_L}$$

$$\frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q}{P \langle \boldsymbol{a}, \boldsymbol{b} \mid \boldsymbol{\alpha}_\pi \rangle Q \vee_0 R}\, {}^{\vee_R} \qquad \frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q}{P \wedge R \langle \boldsymbol{a}_\pi \mid \boldsymbol{\alpha}_\pi, \boldsymbol{\beta} \rangle Q}\, {}^{\wedge_L} \qquad \frac{P \langle \boldsymbol{a}, \boldsymbol{b} \mid \boldsymbol{\alpha} \rangle Q \wedge R}{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha}_p \rangle Q}\, {}^{\wedge_R} \qquad \frac{P \vee_0 R \langle \boldsymbol{a} \mid \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle Q}{P \langle \boldsymbol{a}_p \mid \boldsymbol{\alpha}_p \rangle Q}\, {}^{\vee_L}$$

$$\frac{P \wedge \phi \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle R \quad Q \wedge \neg\phi \langle \boldsymbol{b} \mid \boldsymbol{\beta} \rangle R}{P \vee_\phi Q \langle \lambda \boldsymbol{x}, \boldsymbol{y}.\mathtt{if}(\phi, \boldsymbol{a}\boldsymbol{x}, \boldsymbol{b}\boldsymbol{y}) \mid \boldsymbol{\alpha}_\pi, \boldsymbol{\beta}_\pi \rangle R}\, {}^{cond_L} \qquad \frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q \quad P \langle \boldsymbol{b} \mid \boldsymbol{\beta} \rangle R}{P \langle \boldsymbol{a}, \boldsymbol{b} \mid \lambda \boldsymbol{x}, \boldsymbol{v}, \boldsymbol{w}.\mathtt{if}(|P|^{\boldsymbol{x}}_{\boldsymbol{\alpha x v}}, \boldsymbol{\beta x w}, \boldsymbol{\alpha x v}) \rangle Q \wedge R}\, {}^{cond_R}$$

$$\frac{P \langle \boldsymbol{a}, \boldsymbol{b} \mid \boldsymbol{\alpha} \rangle Q \to R}{P \wedge Q \langle \boldsymbol{a} \mid \boldsymbol{\alpha}, \boldsymbol{b} \rangle R}\, {}^{imp} \qquad \frac{P \wedge Q \langle \boldsymbol{a} \mid \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle R}{P \langle \boldsymbol{a}, \boldsymbol{\beta} \mid \boldsymbol{\alpha} \rangle Q \to R}\, {}^{exp} \qquad \frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q \quad Q \langle \boldsymbol{b} \mid \boldsymbol{\beta} \rangle R}{P \langle \boldsymbol{b} \circ \boldsymbol{a} \mid \boldsymbol{\alpha} *_{\boldsymbol{a}} \boldsymbol{\beta} \rangle R}\, {}^{comp}$$

Quantifier rules: Term introduction, $\boldsymbol{\lambda}$-abstraction and application, epsilon terms.

$$\frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q(\boldsymbol{t})}{P \langle \lambda _.\boldsymbol{t}, \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle \exists \boldsymbol{x}\, Q(\boldsymbol{x})}\, {}^{\exists_R} \qquad \frac{P(\boldsymbol{t}) \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q}{\forall \boldsymbol{x}\, P(\boldsymbol{x}) \langle \lambda \boldsymbol{f}.\boldsymbol{a}(\boldsymbol{f}\boldsymbol{t}) \mid \lambda _.\boldsymbol{t}, \lambda \boldsymbol{f}.\boldsymbol{\alpha}(\boldsymbol{f}\boldsymbol{t}) \rangle Q}\, {}^{\forall_L}$$

$$\frac{P(\boldsymbol{x}) \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q}{\exists \boldsymbol{x}\, P(\boldsymbol{x}) \langle \lambda \boldsymbol{x}.\boldsymbol{a} \mid \lambda \boldsymbol{x}.\boldsymbol{\alpha} \rangle Q}\, {}^{\exists_L} \qquad \frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q(\boldsymbol{x})}{P \langle \lambda \boldsymbol{y}, \boldsymbol{x}.\boldsymbol{a}\boldsymbol{y} \mid \lambda \boldsymbol{y}, \boldsymbol{x}.\boldsymbol{\alpha}\boldsymbol{y} \rangle \forall \boldsymbol{x}\, Q(\boldsymbol{x})}\, {}^{\forall_R} \qquad \boldsymbol{x} \text{ not free in } Q \text{ resp. } P \text{ for } \exists_L \text{ resp. } \forall_R$$

$$\frac{\exists \boldsymbol{x}\, P(\boldsymbol{x}) \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q}{P(\boldsymbol{t}) \langle \boldsymbol{a}\boldsymbol{t} \mid \boldsymbol{\alpha}\boldsymbol{t} \rangle Q}\, {}^{s_L} \qquad \frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle \forall \boldsymbol{x}\, Q(\boldsymbol{x})}{P \langle \lambda \boldsymbol{y}.\boldsymbol{a}\boldsymbol{y}\boldsymbol{t} \mid \lambda \boldsymbol{y}, \boldsymbol{v}.\boldsymbol{\alpha}\boldsymbol{y}\boldsymbol{t}\boldsymbol{v} \rangle Q(\boldsymbol{t})}\, {}^{s_R} \qquad \frac{P_\forall \langle \boldsymbol{a}, \boldsymbol{b} \mid \boldsymbol{\alpha} \rangle \exists \boldsymbol{x}\, Q(\boldsymbol{x})}{P_\forall \langle \boldsymbol{b} \mid \boldsymbol{\alpha} \rangle Q(\boldsymbol{a})}\, {}^{\epsilon_R} \qquad \frac{\forall \boldsymbol{x}\, P_\forall(\boldsymbol{x}) \langle - \mid \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle Q_{qf}}{P_\forall(\boldsymbol{\alpha}) \langle - \mid \boldsymbol{\beta} \rangle Q_{qf}}\, {}^{\epsilon_L}$$

Consequence, extensionality, and induction/recursion

$$\frac{P' \to_D P \quad P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q \quad Q \to_D Q'}{P' \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q'}\, {}^{cons} \qquad \frac{P \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q \quad \boldsymbol{a}, \boldsymbol{\alpha} = \boldsymbol{b}, \boldsymbol{\beta}}{P \langle \boldsymbol{b} \mid \boldsymbol{\beta} \rangle Q}\, {}^{ext} \qquad \frac{P(x) \langle \boldsymbol{a}(x) \mid \boldsymbol{\alpha}(x) \rangle P(x+1)}{P(0) \langle \mathtt{rec}\, \boldsymbol{a} \mid \mathtt{rec}^*\, \boldsymbol{a}\boldsymbol{\alpha} \rangle \forall x\, P(x)}\, {}^{ind}$$